Synthetic sample document - fictional data for demos and evaluation only.

BEARING MAINTENANCE MANUAL
Document ID: MAN-BRG-115
Scope: Deep groove ball bearings and spherical roller bearings on pumps, fans and motors.

1. Installation:
- Heat the bearing with an induction heater to a maximum of 110 degC.
- Never use an open flame to heat a bearing; it destroys the heat treatment.
- Mount the bearing with a fitting sleeve that presses only on the ring being fitted.
- Check radial internal clearance with a feeler gauge after mounting.
- Record the bearing serial number in the asset register.

2. Lubrication:

- Use lithium complex grease, consistency NLGI 2, unless the equipment datasheet says otherwise.
- Relubrication quantity in grams: G = 0.005 x D x B (D = outside diameter, B = width, in mm).
- Over-greasing causes churning and a temperature rise of 10 to 15 degC; stop adding grease when fresh grease appears at the relief port.
- Never mix greases with different thickener types.
- Shielded (2Z) bearings are lubricated for life and must not be regreased.

3. Failure Diagnosis:
- Spalling (flaking) on the raceway indicates rolling contact fatigue at end of life.
- Brinelling (indentations in the raceway) indicates impact or static overload, often during transport.
- Fluting (washboard pattern) is caused by electrical current passing through the bearing; install an insulated bearing or a shaft grounding brush.
- Blue or brown discoloration indicates overheating from lubricant starvation.
- Envelope acceleration above 4 gE requires inspection within one week.